

Outline for *Living Alignment*

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PREVIEW

Section 1

- Living systems are healthier when they avoid fixation on particular modes of interaction (we work through several examples).
- Fixation is limited by semi-permeable boundaries.
- Boundaries translate local optimization pressure into productive effort, contextualizing individual entities as part of larger structures.
- Boundaries yield modularity, deepened individuality, and open-ended discovery. This is what healthy, flourishing living systems do.

Section 2

- Although AI alignment has many definitions, one thing most people agree on is that an aligned future is one full of health and flourishing.
- Therefore, alignment involves ongoing construction of boundaries at every scale, to avoid fixation.
- No fixed alignment scheme can be aligned, and ongoing discovery is intrinsic to alignment.

Section 1 – Health in living systems

Intro

- Living systems are healthier when they avoid fixating on, or reducing to, particular patterns of interaction. Examples: a single drive (food, sex) pursued exclusively becomes pathological; monoculture is fragile; cancer; seizure; obsession; monopoly; one person's will restricting many.
- The systems that succeed and persist are those with barriers on their own myopic drives. E.g., precommitment against future impulsivity, plant self-incompatibility, the genome constraining a selfish gene. (In turn, that whole system is a part in another system, and is similarly bounded.) These barrier systems have been fine-tuned by what actually works across long evolutionary history.
- Effective barriers are semi-permeable, permitting conditionally stable interactions without letting them become rigid or dominating. A well-functioning barrier does not shut down a drive, but contextualizes it within a larger system.
- Healthy living systems are generative. Barriers form modular, non-overfit solutions. These become the engine of discovery.

Example: genomes

- Sex is expensive: mate search, half the population not bearing offspring, meiosis halving each parent's genetic representation. Yet nearly all eukaryotes are sexual, and many hermaphrodites go to great lengths to prevent self-fertilization. Why place a barrier on the path to reproduction?
- Asexual lineages lock genes together. Selection cannot act on one without dragging the rest, deleterious mutations are permanent (Muller's ratchet), and beneficial mutations in different individuals can never combine. Alleles overfit a fixed genetic background.
- Recombination breaks associations among loci, letting selection act more independently and exposing alleles to varying backgrounds. Favors alleles that perform well across contexts. The result is modularity, where each gene becomes an interchangeable unit that can contribute something meaningful to many different genetic contexts. Which creates the foundation for faster discovery of new solutions.

Example: goals

- Nutrition, osmotic balance, temperature, reproduction, pain avoidance — each evolved as an imperfect proxy for fitness, so fixation on any one paradoxically reduces fitness.
- Same for higher-order goals and strategies. E.g. excess focus on work leads to burnout. Example of Hershberger's chickens: a food cup was rigged to retreat at twice the chicken's approach speed, so food was obtainable only by running away. The chickens were defeated by their own prepotent approach strategy.
- Cognitive control as an example of semi-permeable boundary. Makes a drive's expression conditional rather than nullifying it. Control holds context, rules, and distal goals that bias competition among representations. Hunger guides eating without overeating.
- Each drive carries a compressed picture of the world and favors the congruent path under that partial representation. A barrier makes the congruent path unavailable and forces symmetry breaking (eg, move away from the food). Stymying a shallow objective often produces progress on a deeper one, driving discovery.

Example: frames

- Every perspective is partial. From inside a frame, the frame is invisible and simply looks like reality.
- Losing the ability to move between frames is a signature of psychopathology: rigid negative beliefs in depression and anxiety; intrusive thoughts in obsessional disorders; delusions held with conviction against evidence in psychosis.
- Awareness: transition from fixed identification with a frame (subject) to seeing that frame as just another content of experience (object). Awareness limits fixation on any particular belief or mental form. Becoming aware of a frame doesn't make it wrong; it's held for its partial truth. Holding many partial truths in delicate balance activates deeper sensitivity and generativity.
- Kuhn and Popper. Over-commitment to theories gets science stuck. On the other hand, scientists should resist change to a degree. The point is not to shut down myopic perspectives. The point is to contextualize each frame so it responds to evidence. Healthy science turns anomalies into crises and revolutions—discovery.

Example: groups

- The greatest achievements of humans depend on structured interaction among individuals with distinct points of view. Barriers support individuals developing their own perspectives, but because they are semi-permeable, people can also combine solutions.
- Example of insufficient barriers. In Zollman's peptic-ulcer case, one overly influential result propagated through the field and medicine settled on a wrong understanding for decades.
- Bernstein et al.: groups solving traveling-salesman problems did better with intermittent rather than continuous information exchange. During periods of independence, members pursued diverse solutions instead of getting stuck on seductive dead ends inherited from other people.
- Craven's search for the USS Scorpion. Experts used their distinct expertise, never consulting each other. Aggregated prediction landed within 300 yards of the vessel.
- The discovery of the printing press arose from Gutenberg combining the steel punch (developed by goldsmiths) and the screw press (developed by farmers). A group with many ideas in circulation, each developed through the unique pressures of semi-isolation, is ripe for such recombination.

Example: institutions

- Every actor is myopic: self-interested, future-discounting, or attached to a perspective. Unchecked optimization from one actor's point of view is typically harmful. For example, profit motives cause deception and exploitation; dominance motives can cause silencing of others; even prosocial beliefs

can cause conflict when groups see differently. Individual perspectives must be contextualized.

- Semi-permeable boundaries: constitutions, laws, norms, courts, regulators. Their purpose is not to nullify individual intentions but to contextualize them. This productively reroutes energy. For example, a profit motive leads to innovation and new discoveries.
- However, intelligent agents don't accept the limits set on their desires, so institutions adapt as loopholes are found, gaining robustness by being tested against many situations and motives.

Other examples

- Cell membranes with conditional gates put gradients to work that would otherwise destroy the cell.
- Symbiogenesis. Archaea and protobacteria evolving separately for hundreds of millions of years, finding distinct solutions in relative isolation. The eukaryotic cell was discovered by combining these solutions.
- Brains spend huge resources keeping representations separate while letting them interact (lateral inhibition, pattern separation). Healthy brain dynamics live between pure synchrony and uncorrelated noise, with pathology like Parkinson's in excess correlation. Imagination and discovery through structured synthesis (as in replay).
- Psychological boundaries in relationships.
- Each entity in an ecosystem is held in check by boundaries like predation and competition. When these fail, as with invasive species, ecosystems collapse. Ecological relationships create new niches and new functions.

Section 2 – The alignment problem

Intro

- In the previous section, we suggested a general way to think about what health and flourishing mean, derived from the study of living systems. Here we apply this framework to the AI alignment problem.
- We propose that an aligned future has the generativity, robustness and modularity of life at all scales.
- These properties depend on evolving networks of semi-permeable barriers which contextualize new myopic processes as they arise.
- This notion of alignment may be suited for reasoning about futures that look very different from the present.

Reward hacking

- Whatever fixed value function you write down is not 'what you really want' in a deeper sense.
- Example of RLHF learning sycophancy. Other examples of reward hacking.
- Our real values are incredibly complex. Sugar on the tongue vs long-term health; grandchildren and strangers abroad and planetary sustainability; different values at different times and across people; deliberated values differing from snap ones; values that resist verbal expression, reaching below language into bodily, communal, environmental intuition; values that evolve as we learn. Any expression of our values is incomplete, like a planarian lacking the concepts to entertain human values.
- So if a powerful system does a good enough job optimizing for any given value function, the results are harmful. Fixation on one way of specifying values is insensitive both to existing structure it fails to capture and to structure that does not yet exist.

Inverse methods

- Consider the example of inverse methods.

- There are advanced alignment methods that treat the value function as uncertain and continually updated, so an agent unsure of human wants becomes risk-averse and correctable. It potentially also accommodates values changing over time.
- However, any fixed system for learning values still rests on assumptions: what counts as human behavior, what counts as a human, how behavior maps to values, the formalism for describing values, how uncertainty is represented and updated, how value becomes action.
- Fixing those concepts is misaligned in the long run.

No scheme is alignment

- More generally, any form, pattern, or conceptual scheme is misaligned if taken too seriously. Therefore no fixed approach can achieve alignment.
- This is the lesson of life: healthy living systems are endlessly generative with many scales of partially separated perspectives loosely and modularly coupled.
- Importantly, this does not mean having no scheme. Discarding schemes capriciously loses as much subtlety as any other fixation. Existing AI safety approaches are necessary, and discovering new schemes and formalizations is essential to progress.

From alignment to living alignment

- Conclusion from above: it's not enough to simply 'do what's good for humans' in a naive sense. That notion breaks down.
- Things will change, including what there is to value and how it is valued. Think of the world today compared to the world a billion years ago. Any formalism or concept we have now will eventually appear as limited as a single-celled organism's notions of 'what is good and bad'.
- Therefore, the problem of alignment, if carefully inspected, transforms into the open-ended problem of living alignment.
- This means life-like systems that open-endedly discover new ways of contextualizing any myopic processes. Preserving and extending the richness and generativity of life.

Is AI special?

- With AI, things change faster, and there is a lot of power behind fixed forms. In the past, there have always been arms races between different patterns of organization, and sometimes one gets so far ahead that the local area is decimated. But if a version of AI gets so far ahead of everything else, there's the possibility of world-level destruction.
- AI systems increasingly are acting autonomously, faster than humans can follow, in ways humans cannot understand.
- Humans have always found and solved problems. But the iterative process may not keep working: AI already shows rapid global adoption, intensive resource use, concentrated information flow, anthropomorphization, wholesale offloading of cognition, and rapid improvement; and may show superhuman intelligence and recursive self-improvement without a restrictive biological substrate.
- Example: concentration of power. Human power amplified by persuasion, autonomous weapons, surveillance, falling demand for human labor.
- Example: conceptual monoculture. Loss of diversity in beliefs, frames, values, and problem-solving approaches. This diversity exists within-person, between people, between humans and AI, and between organizations, fields, cultures, nations.

Healthy for whom?

- Our framework defines aligned systems as those with lifelike properties of multi-scale agentic modularity, which avoid collapse onto fixated forms.
- But alignment is typically concerned with human interests. What happens if AI replaces humans

with its own lifelike systems, but destroys humanity in the process?

- Eventually, if there are descendents of humans, they will presumably be drastically different from what we look like at present. Holding things in stasis eternally is not what we want.
- However, in the short to medium term, honoring existing life and values is where it's at. It is difficult to overstate the amount of complexity and content in existing life, including in the relationships between organisms and in the mental lives of humans. Existing life has been through so much to get here (billions of years with parallel experiments all over Earth).

Going forward

- The industry is already building in internal barriers. Some of these have humans in the loop to greater or lesser degrees, like governmental oversight, mechanistic interpretability, scalable oversight. Some are fully automated (automated CoT monitoring, agentic red teaming, guardrail models etc).
- Again, any fixed conceptual approach is misaligned.
- The essence of living alignment is transitioning to a system (which includes humans, AI, as well as other species and ecosystems on earth) that carries forward the full momentum and generativity of life, rather than fixating.
- The coarsest version of fixation are obvious: e.g., paperclips, bioweapons, extreme concentration of power.
- What we have to be careful about are less-obvious versions. A subtle fixation at first appears harmless, but the system as a whole is collapsing toward something without contextualization. AI erodes boundaries past a healthy balance. Conceptual monoculture leads to an impoverished future, or loss of drive. Systems becoming more correlated, and therefore less resilient to shocks and less capable of future discovery.
- On the other hand, a bright future is also possible. AI creates more structure in the world, including new boundaries to contextualize its own new abilities. AI tools for bounding AI (finding bugs before exploitation, detecting deepfakes, scalable oversight); well-tuned AI nurturing individual perspective and curiosity; democratic AI helping different perspectives coexist and cooperate; new niches created faster than old ones are destroyed; room on Earth for increasing diversity.
- The good version increasingly looks like subject-to-object perspective-taking within AI and human-AI systems. Stepping back to hold a previously axiomatic thing in a larger context. Internal discovery within systems that represent themselves and the world.
- We suspect that both outcomes are possible and therefore it is critical to act wisely now.
- Of course, the ideas in this paper come with the same proviso that we shouldn't over-attach to them.

Conclusion

- Health and flourishing are not any particular form or concept. Therefore, alignment is not picking the right values or principles, nor the right system for learning them, nor any method for interpretability or corrigibility. All can be useful parts of alignment.
- Instead, aligned AI continues life's process of maintaining and enhancing the individuality of distinct elements while also deepening their relatedness. Distinct elements experiment with their own solutions from their own perspectives, expanding a reservoir of uniquely useful components for the larger systems containing them. Every entity is inevitably myopic, yet the whole unfolds robustly because different perspectives reroute each other's sharp edges as constructive momentum.
- By more fully respecting the staggering subtlety of the existing world, aligned AI supports the open-ended discovery of new depth.